Distribution shape adds structure to location and spread

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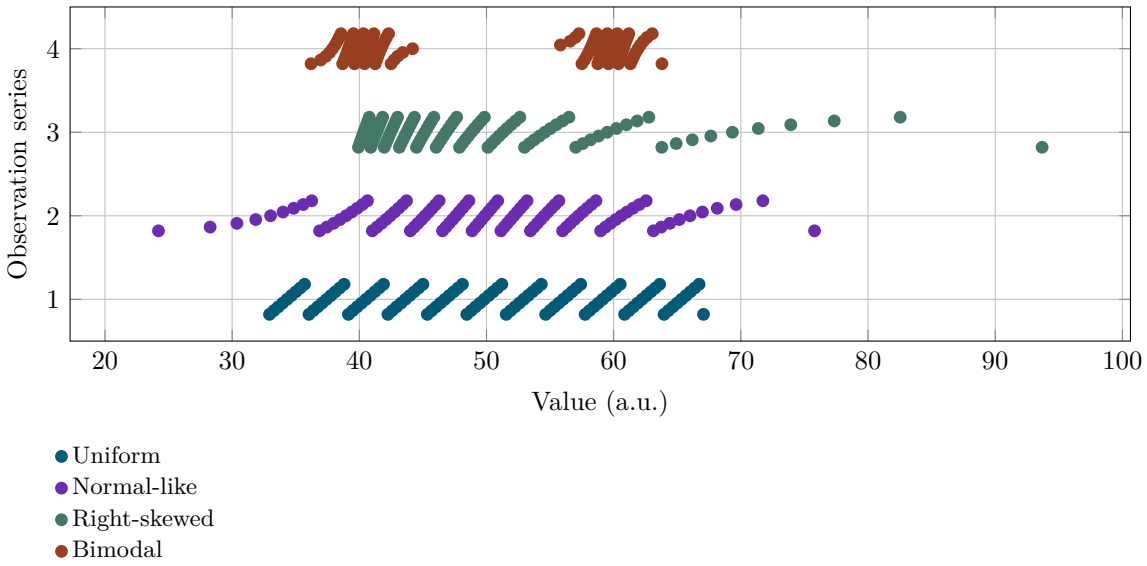
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1 Ordering observations reveals where the data accumulate

Location and spread each compress a collection into a small set of numbers. The comparison in Fig. 1 shows why we also inspect the arrangement of the observations: different patterns can share similar summaries. Grouping, asymmetry and long tails carry information about the process that produced the measurements.

Fig. 1. Authored data patterns illustrating the additional information carried by distribution shape.



The data notation in Eq. 1 identifies the sample, and the order-statistic notation in Eq. 2 assigns ranks after sorting. The range (Eq. 3) describes the distance between the observed endpoints. Because those endpoints are individual observations, their behaviour depends strongly on sample size and on extreme values. This makes the range useful as an observed extent with an explicitly stated sampling context.

Eq. 1. Sample notation with observation identity, sample size and quantity conventions.

$$x_1, \ldots, x_n, \qquad n = 5$$
$$(10.2,\ 9.8,\ 10.0,\ 10.0,\ 10.1)\ \text{mg/L}$$

Symbols
What does i identify? i labels the observation in acquisition order, not its rank. n counts all readings, including ties. Synthetic teaching data.

Eq. 2. Ordered-observation notation with rank and tie conventions explained.

$$(x_1, \ldots, x_5) = (10.2, 9.8, 10.0, 10.0, 10.1)$$
$$x_{(1)} \leq \cdots \leq x_{(5)}$$
$$(x_{(1)}, \ldots, x_{(5)}) = (9.8, 10.0, 10.0, 10.1, 10.2)$$

Symbols
Same values, different meaning of the index All values are in mg/L. x_1=10.2, but $x_{(1)}$=9.8 is the minimum. $x_{(2)}$=$x_{(3)}$=10.0: two observations, not one. $x_{(n)}$ is the maximum.

Eq. 3. Observed range as the difference between the largest and smallest values.

$$R = x_{(n)} - x_{(1)}, \qquad n \geq 1$$

Symbols
Order first $x_{(1)}$ is the minimum; $x_{(n)}$ is the maximum. R is a width, not the endpoint pair returned by R's range(x).

Quartiles and other quantiles describe locations within the ordered collection. The quartile construction in Eq. 4 and the interpolation rule in Eq. 5 make the chosen finite-sample convention reproducible. The application in Fig. 2 connects a selected probability level with a concentration threshold. Reporting the convention is particularly useful for small samples, where alternative accepted definitions can give visibly different answers.

Eq. 4. Authored quartile-position convention and its interpolation notation.

$$Q_1 = \widehat{Q}(0.25), \quad Q_2 = \widehat{Q}(0.50), \quad Q_3 = \widehat{Q}(0.75)$$

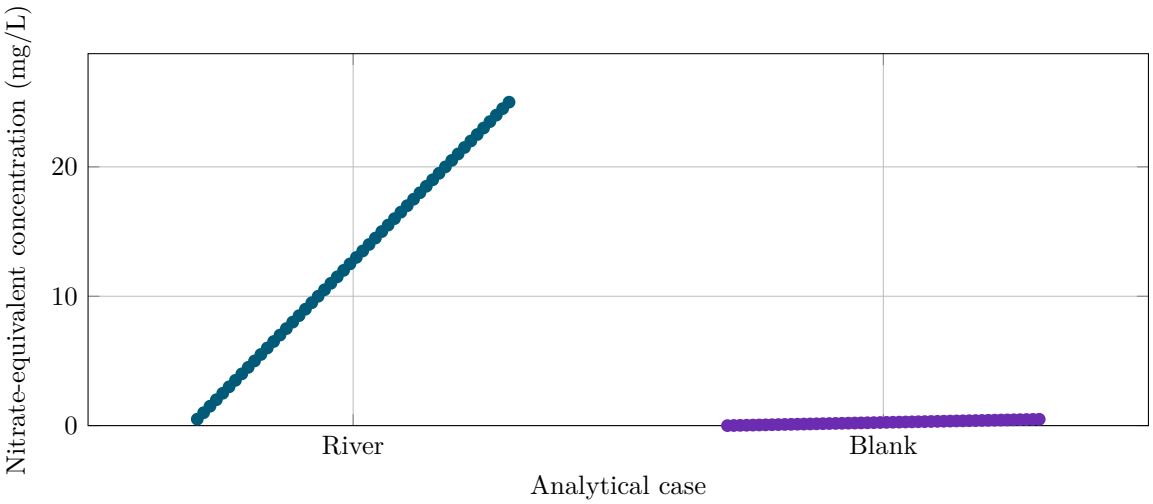
Symbols
Quarter, half, three quarters Q_1, Q_2 and Q_3 are the lower quartile, sample median and upper quartile. Q(p) is the linearly interpolated sample quantile at cumulative proportion p. The fractions 0.25, 0.50 and 0.75 are unitless; all quantile values have the measurement unit.

Eq. 5. Finite-sample quantile interpolation with the complete probability, rank and endpoint conventions.

$$h = 1 + (n-1)p, \quad j = \lfloor h \rfloor, \quad d = h - j$$
$$\widehat{Q}(p) = (1-d)x_{(j)} + d\,x_{(j+1)}$$
$$n \geq 2, \quad 0 \leq p < 1$$

Symbols
Interpolate between adjacent ranks p: unitless cumulative proportion; n: number of observations; h: fractional rank; floor(h): round h down to integer j; d: unitless interpolation weight h−j. $x_{(j)}$ and $x_{(j+1)}$ are adjacent ordered values; Q(p) is their weighted average, in the measurement unit. The factors 1−d and d sum to one. The displayed interior case requires n≥2 and p<1; at p=1 return the maximum, and for n=1 return the sole observation.

Fig. 2. Authored quantile application linking an ordered concentration sample with selected cumulative levels.



2 Choose a spread summary for the part of the distribution of interest

A spread measure emphasizes particular features of the observations. The interquartile range in Eq. 6 measures the distance from the lower quartile to the upper quartile. Its application in Fig. 3 describes the central half of the ordered data using the authored quantile convention. The tail comparison in Fig. 4 shows why we retain tail observations alongside this central summary.

Eq. 6. Interquartile range in the original measurement unit, using the authored quartile convention.

$$\mathrm{IQR}(x_1, \ldots, x_n) = \widehat{Q}(0.75) - \widehat{Q}(0.25), \quad n \geq 1$$

Symbols
A width in the original unit x_1,...,x_n: observed values; n≥1: sample size. Q(p): linearly interpolated sample quantile at unitless proportion p. Subtract the lower quartile at 0.25 from the upper quartile at 0.75. IQR is this central width in the measurement unit, not the entire observed range.

Fig. 3. Authored IQR application showing the separation of the lower and upper quartiles.

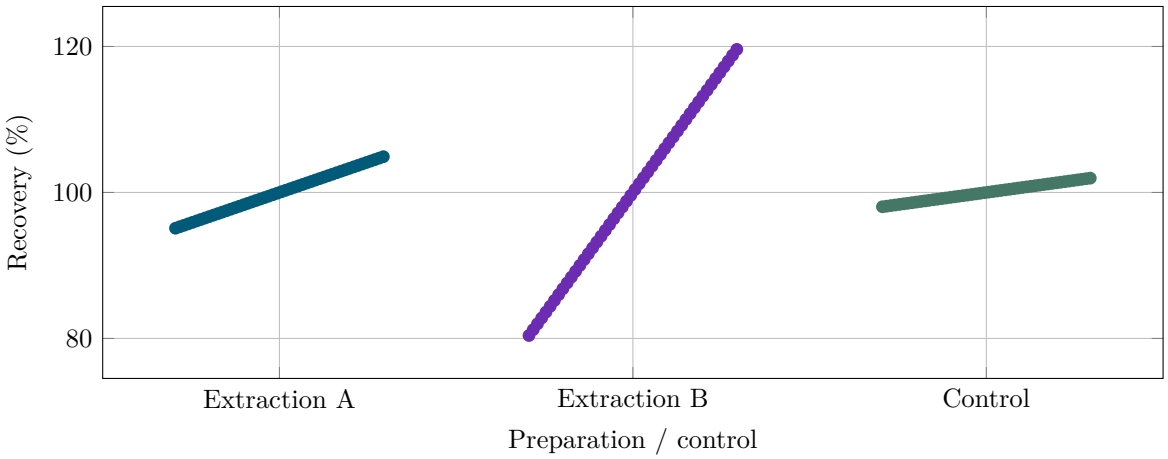
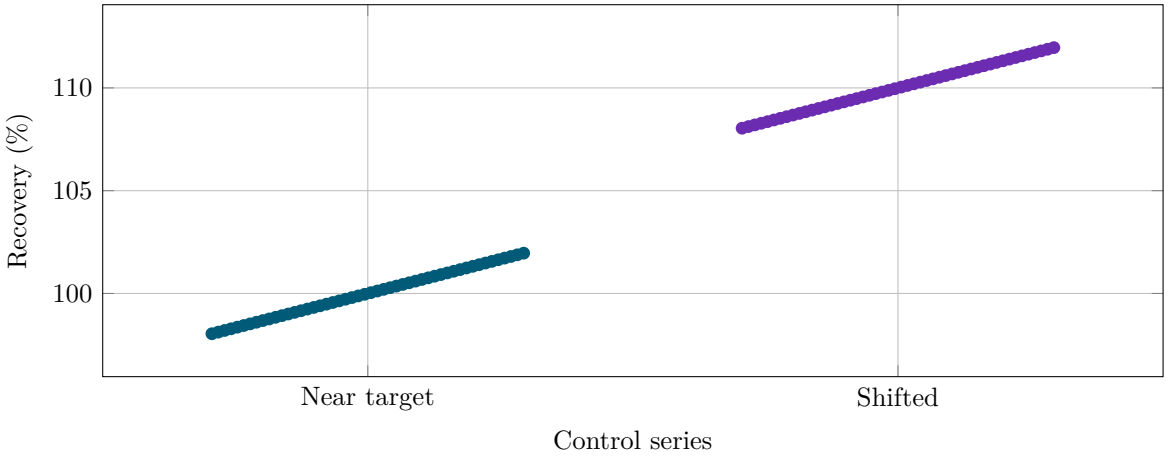


Fig. 4. Tail comparison accompanying the IQR: central quartile separation is read together with the extreme observations.



The median absolute deviation first measures absolute distance from the median and then takes the median of those distances (Eq. 7). The worked application in Fig. 5 makes this two-stage ordering explicit. The raw quantity has the original measurement unit. A scaled version additionally specifies the scaling constant and its reference distribution, so the reported name and convention identify which quantity was calculated.

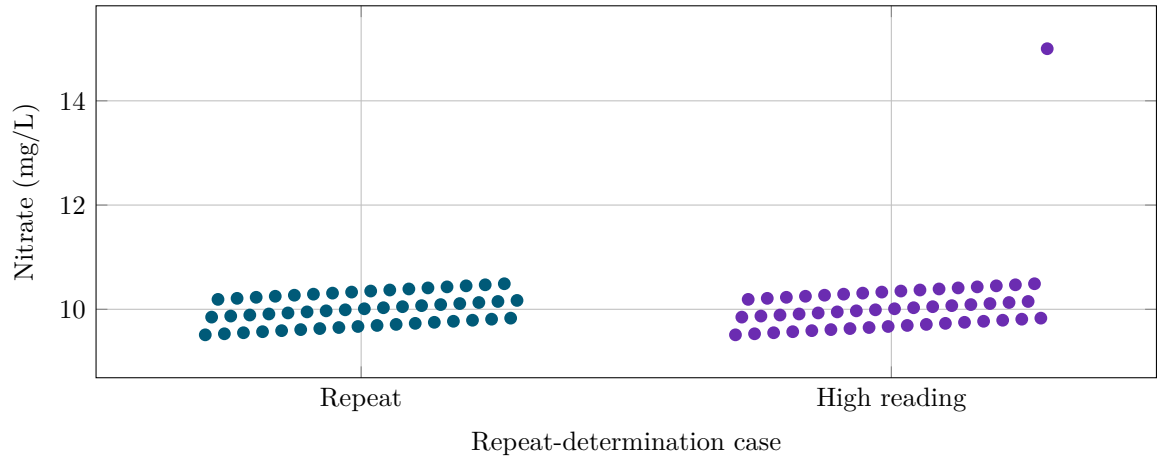
Eq. 7. Median absolute deviation with its centring, absolute-value and scaling conventions.

$$\mathrm{MAD}_{\mathrm{raw}} = \mathrm{median}_i \left| x_i - \mathrm{median}_j(x_j) \right|$$

Symbols

Two median operations First find the sample median; then find the median of all absolute distances to it. No squaring and no averaging of these distances.

Fig. 5. Authored MAD example illustrating absolute distances from the sample median.



Comparisons under the authored Normal and skewed examples (Fig. 6, Fig. 7) connect robust central spread with the overall shape. Standard deviation emphasizes squared departures; IQR emphasizes quartile separation; MAD emphasizes the typical absolute departure from the median. Choosing among them starts with the analytical question and the part of the distribution that carries the relevant information.

Fig. 6. Spread summaries in the authored Normal reference example.

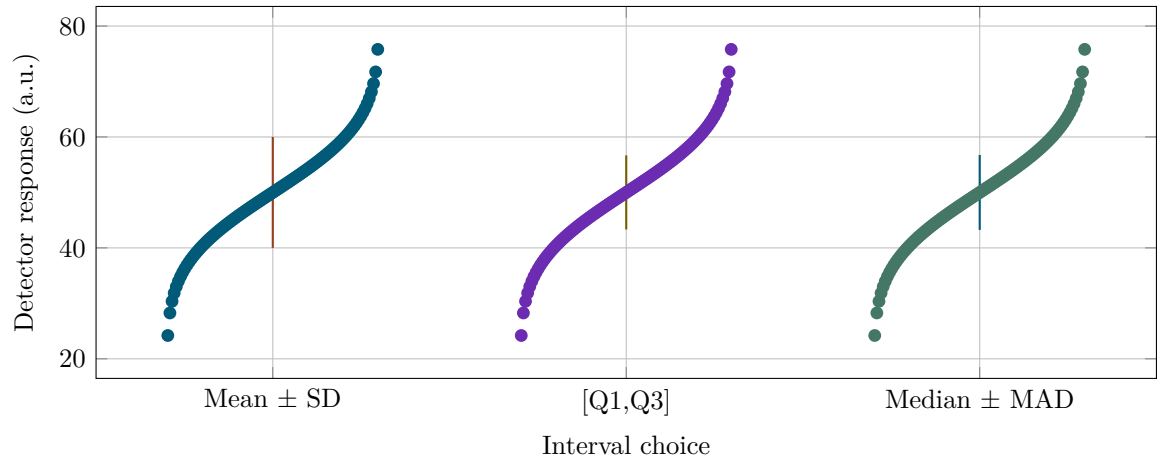
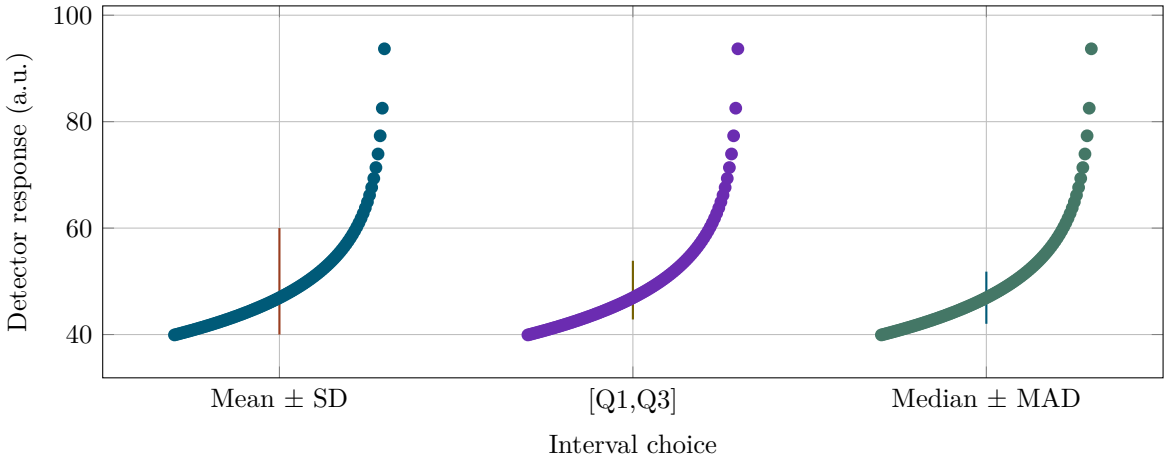


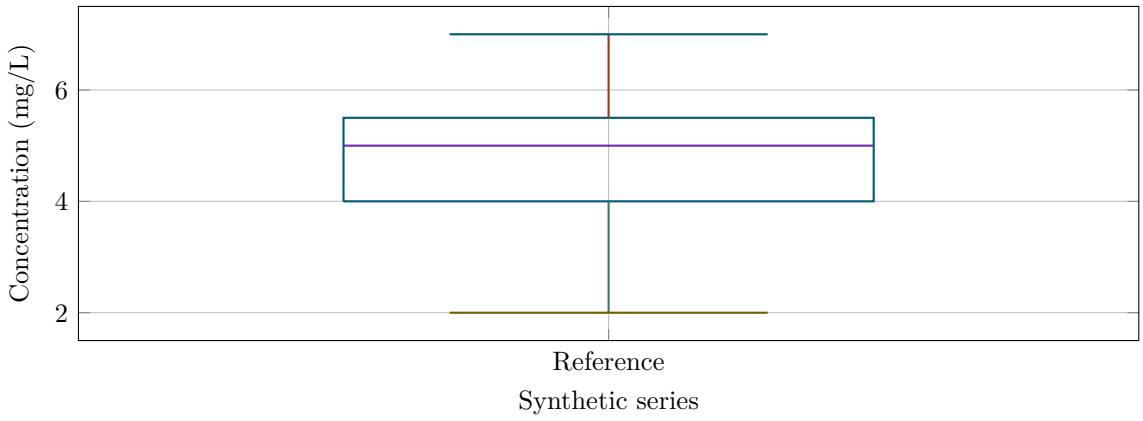
Fig. 7. Spread summaries in the authored skewed example, with the asymmetric distribution retained.



3 A boxplot compresses ordered structure into a comparison view

A boxplot places the median and quartiles on the measurement axis. Its box therefore provides a compact view of location and central spread. The construction in Fig. 8 connects the ordered observations with those graphical elements. Whiskers and individually displayed extreme points follow the authored convention, which belongs with the plot's interpretation.

Fig. 8. Boxplot construction from the authored ordered observations, quartiles and whisker convention.



The grouped comparison in Fig. 9 uses this compression to compare several data sets on a common scale. The extreme-value view in Fig. 10 keeps unusual observations visible. A flagged observation invites investigation of its measurement and sampling context; the flag is an operational feature of the plotting convention.

Fig. 9. Authored grouped boxplot application on a common measurement scale.

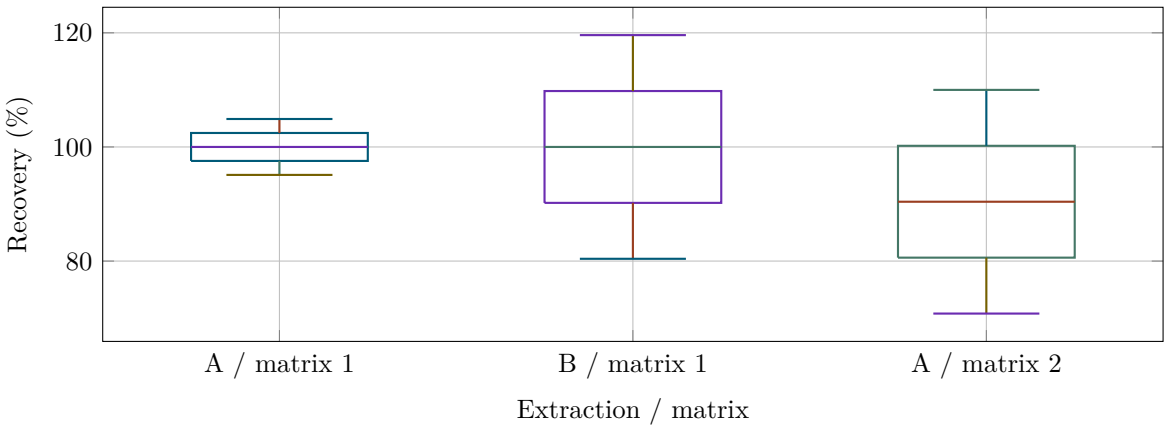
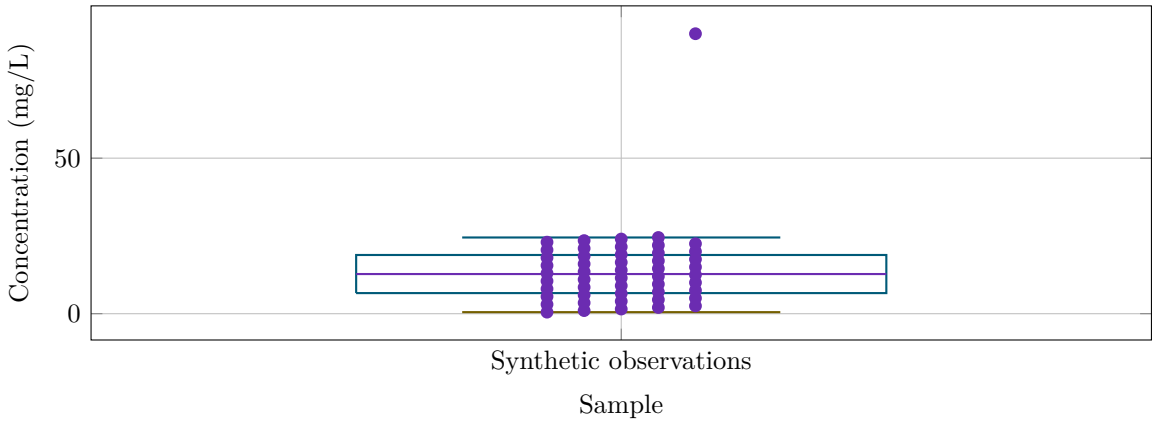
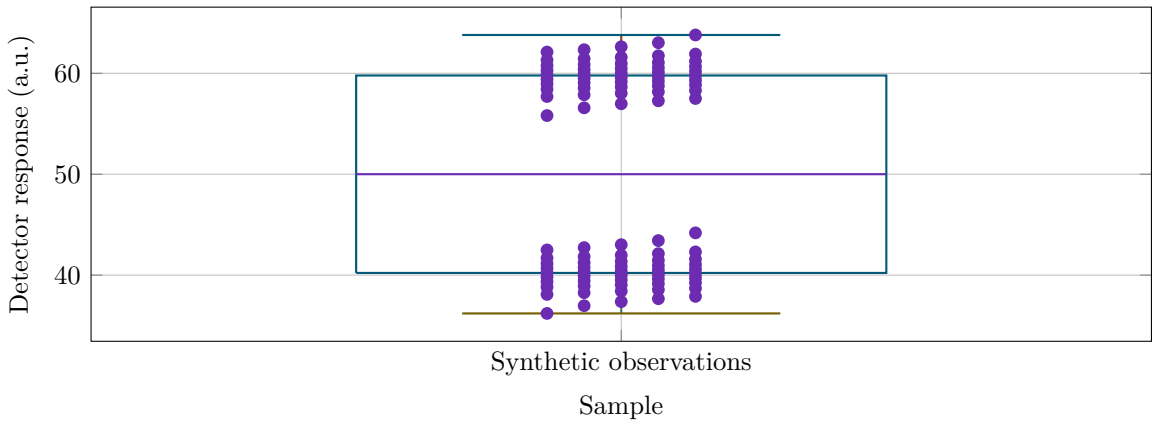


Fig. 10. Extreme observations displayed according to the authored boxplot convention.



Compression also motivates a complementary view. The bimodal example in Fig. 11 retains its underlying observations so that grouping within the box can be examined. Dot plots or histograms can show that internal structure directly. The optional source in Listing 1 allows the same data and plotting convention to be reproduced and compared with fuller distribution views.

Fig. 11. Boxplot and underlying bimodal observations, retaining the two-group structure for comparison.



Listing 1. Optional R source for reproducing the boxplot data and comparison views.

```
if (!requireNamespace("ggplot2", quietly = TRUE)) {
  if (requireNamespace("webr", quietly = TRUE)) webr::install("ggplot2")
  else stop("Install ggplot2 first: install.packages('ggplot2')")
}
library(ggplot2)
groups <- c("A / matrix 1", "B / matrix 1", "A / matrix 2")
d <- data.frame(series = rep(groups, each = 50),
  value = c(seq(95.1, by=0.2, length.out=50),
            seq(80.4, by=0.8, length.out=50),
            seq(70.8, by=0.8, length.out=50)))
d$value[150] <- 110
d$series <- factor(d$series, levels = groups)
p <- ggplot(d, aes(series, value)) +
  geom_boxplot(coef = 1.5) +
  labs(x = "Synthetic extraction / matrix", y = "Recovery (%)") +
```

```
  theme_minimal(base_size = 16)
print(p)
ggplot_build(p)$data[[1]][c("lower", "middle", "upper", "ymin", "ymax", "outliers")]
```

ggplot() + aes()

Map the extraction/matrix group to x and recovery (%) to y.

geom_boxplot(coef=1.5)

Use ggplot2's usual linearly interpolated quartiles and 1.5×IQR fences; flag rather than remove distant values.

ggplot_build(p)$data[[1]]

Inspect the computed quartiles, whisker endpoints and separately plotted observations.

c(...)

Combine values or output-column names into a vector; does not sort them.

data.frame(...)

Build a table with one row per synthetic observation and named series/value columns.

seq(...)

Generate a synthetic grid: **by** sets its increment; **length.out** sets the number of values.

rep(groups, each=50)

Repeat every group label **50 times**, aligning labels with the synthetic observations.

factor(..., levels=groups)

Declare categories and their display order; the recovery values do not change.

requireNamespace(..., quietly=TRUE)

Check package availability without attaching it. TRUE/FALSE selects the setup branch.

webr::install("ggplot2")

Download ggplot2 into browser R if missing. **Network required**; learner observations are not uploaded.

stop(...)

Stop with a setup error. Desktop R needs ggplot2 installed separately.

library(ggplot2)

Attach the installed package so its functions can be called by name.

print(p)

Draw the ggplot object explicitly in the browser R runtime.

labs(x=..., y=...)

Label axes, including the **recovery unit (%)**.

theme_minimal(base_size=16)

Set the plot theme and base font size; **appearance only**, not a statistical calculation.

<-, $, [, [[, +, if / else, ! and ::

<- assigns; **$** selects a column; [subsets; [[extracts a list element. + adds ggplot layers; **if/else** branches; **!** negates; **::** selects a package function.

4 Histogram bins determine the resolution of the observed shape

A histogram groups numerical observations into intervals. The authored Normal, skewed and bimodal examples (Fig. 12, Fig. 13, Fig. 14) show how the resulting bars describe concentration, asymmetry and grouping. Each view is an empirical summary at a selected resolution.

Fig. 12. Histogram of the authored Normal example with its fixed bins and axis normalization.

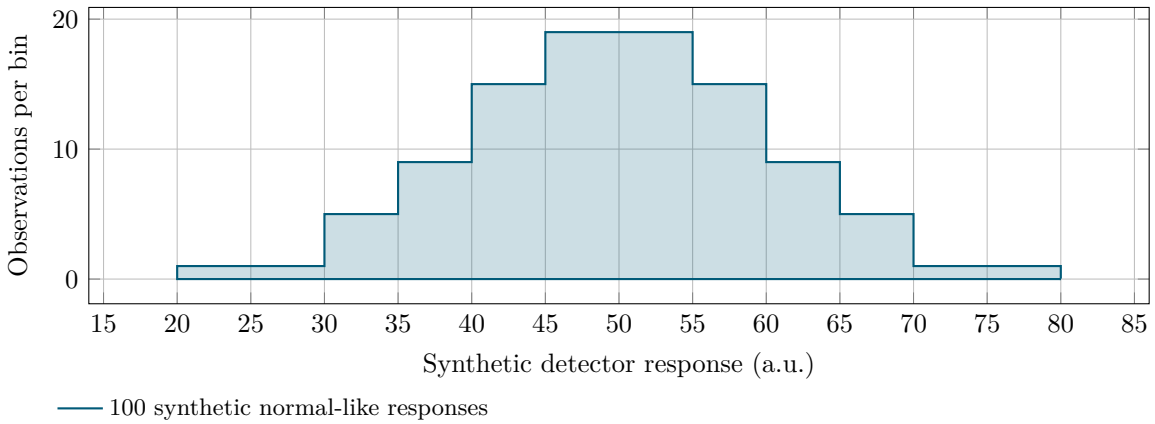


Fig. 13. Histogram of the authored right-skewed observations.

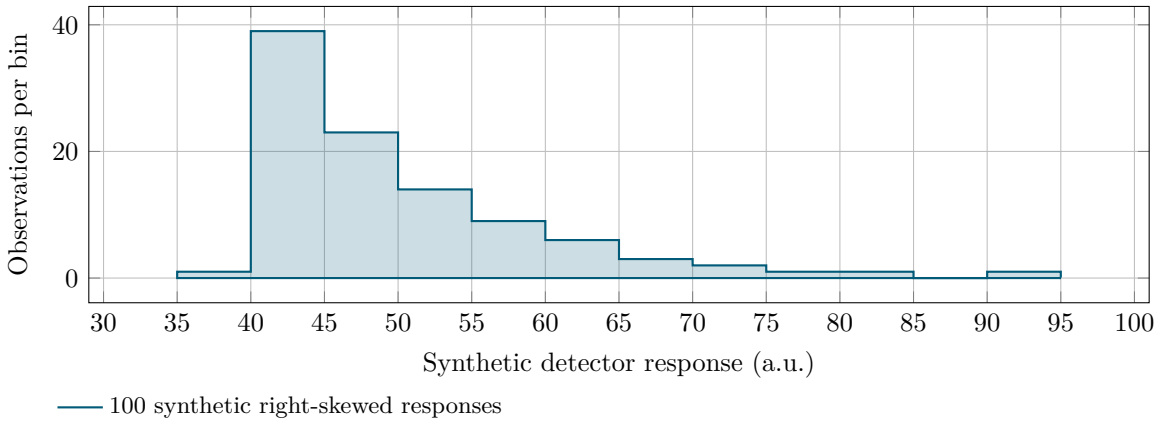
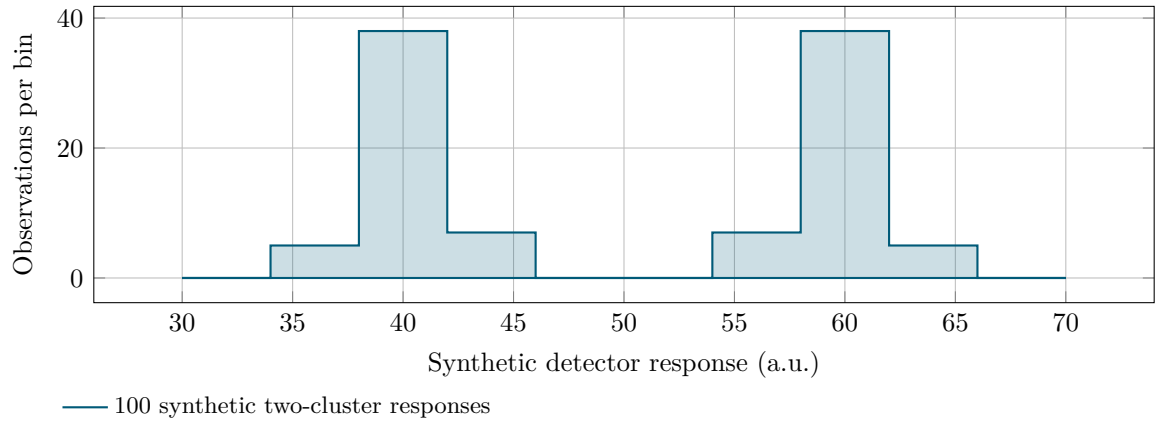


Fig. 14. Histogram of the authored bimodal observations at the selected reference binning.



Bin width changes that resolution. The coarse and fine views in Fig. 15 and Fig. 16 trade broad smoothing against sensitivity to small-scale sampling variation. The origin comparison in Fig. 17 adds a second choice: moving the interval boundaries changes which nearby observations share a bin. Comparing several reasonable settings helps distinguish stable structure from the particular partition.

Fig. 15. Coarse-bin view of the authored comparison data.

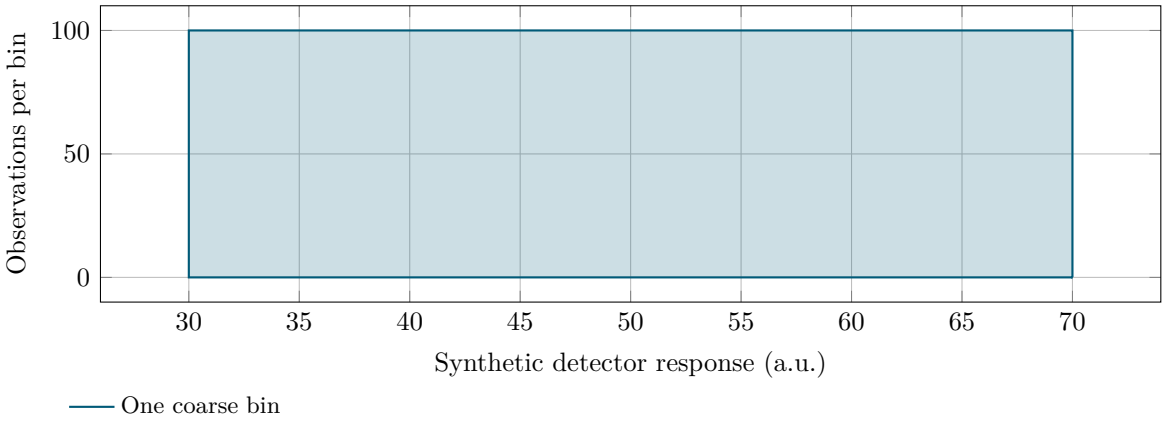


Fig. 16. Fine-bin view of the same authored comparison data.

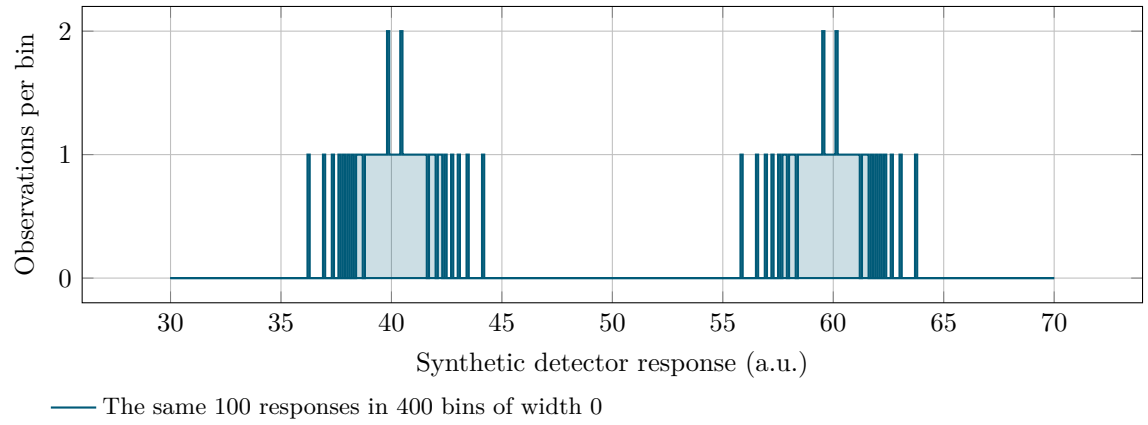
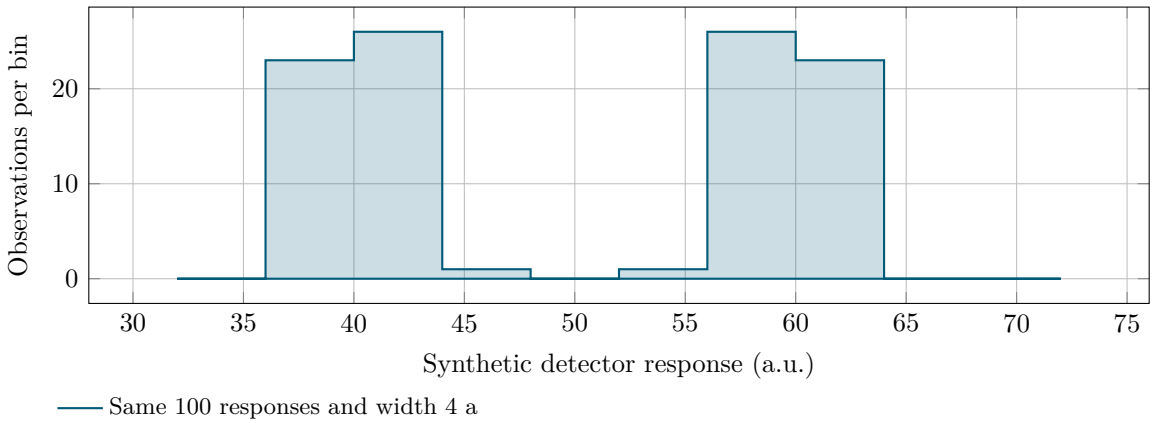


Fig. 17. Histogram-origin comparison retaining the same observations while shifting bin boundaries.



The rules in Eq. 8 provide reproducible starting widths, using the sample size and the authored spread summaries. The adjustable example in Fig. 18 supports checking those choices against the observations. A reported histogram therefore identifies the quantity, units, vertical-axis normalization and interval construction. Its interpretation can then focus on features that remain meaningful at the chosen resolution.

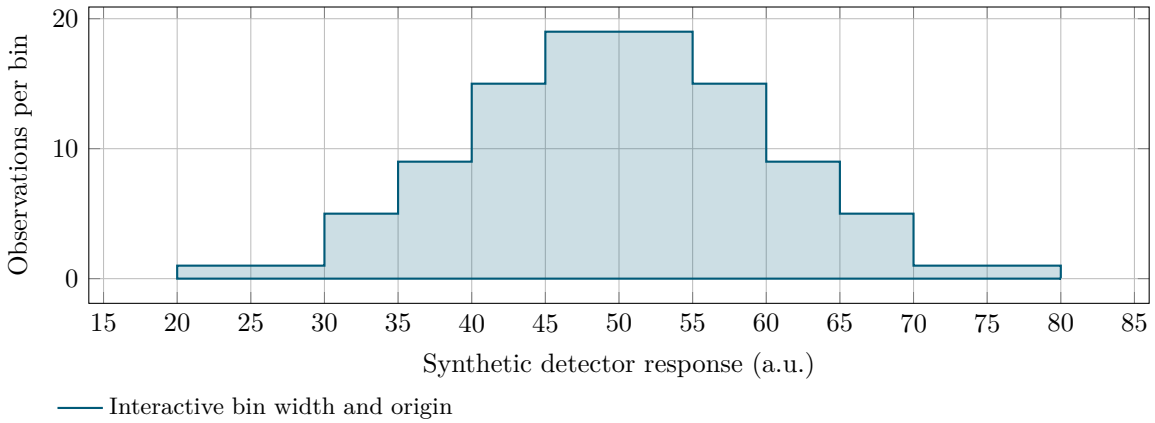
Eq. 8. Authored histogram-width rules with sample-size, spread and unit conventions.

$$h_{FD} = 2\,\mathrm{IQR}\,n^{-1/3}$$
$$h_{Scott} = 3.5\,s\,n^{-1/3}$$
$$k_{Sturges} = \lceil 1 + \log_2 n \rceil$$

Symbols

Names and symbols FD = Freedman–Diaconis; Scott = Scott's rule; Sturges = Sturges' rule. h: width; k: bin count; n: observations; s: sample SD. IQR uses linear interpolation.

Fig. 18. Adjustable histogram at its authored reference settings; the interactive view retains the available parameters.



5 For unequal widths, area carries the relative frequency

Unequal-width intervals make the distinction between bar height and bar area essential. The density-height formula in Eq. 9 divides relative frequency by interval width. Multiplying that height by its width recovers the fraction of observations in the bin. Density consequently has reciprocal measurement units, while relative frequency is dimensionless.

Eq. 9. Density normalization for unequal-width histogram intervals: bar area equals observed relative frequency.

$$h_j = b_{j+1} - b_j$$
$$d_j = \frac{n_j}{n\, h_j}$$
$$d_j h_j = \frac{n_j}{n}, \qquad \sum_j d_j h_j = 1$$

Symbols
Reading the symbols b: edges; h_j: bin width; n_j: count; n: total. Density has inverse measurement units; bar area is the sample fraction.

The unequal-bin construction in Fig. 19 and the adaptive example in Fig. 20 retain the authored boundaries and observations. A narrow interval containing the same fraction as a wider interval requires a taller density bar. This relationship explains why reading the vertical-axis label is part of reading the distribution.

Fig. 19. Authored unequal-width bins with density heights corresponding to their observed fractions.

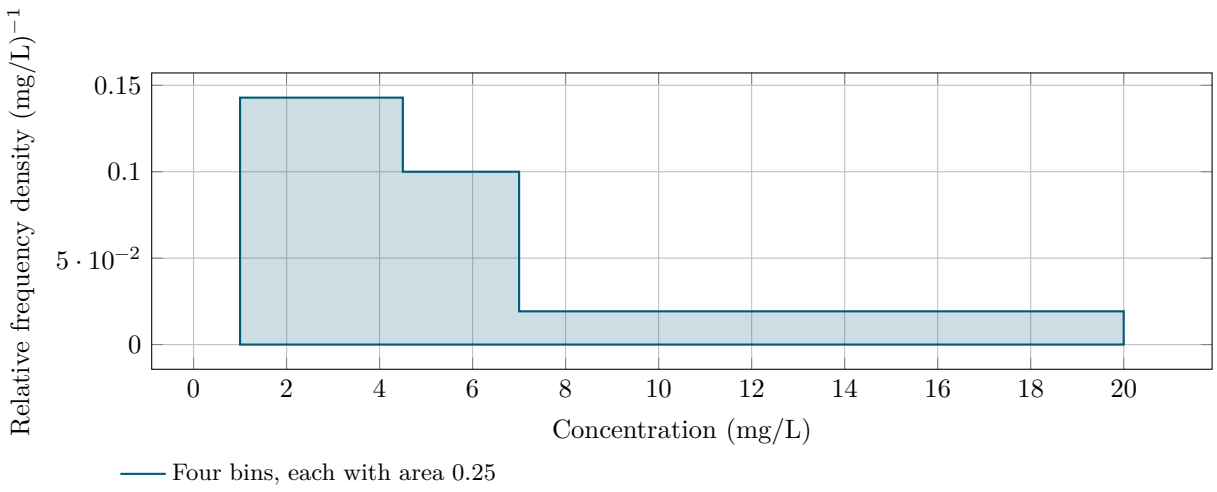
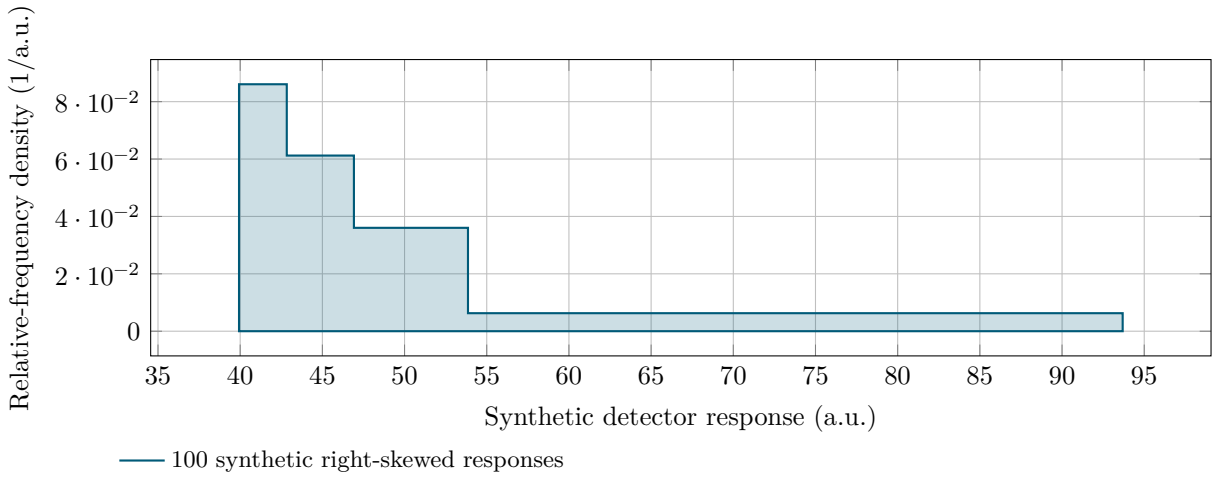
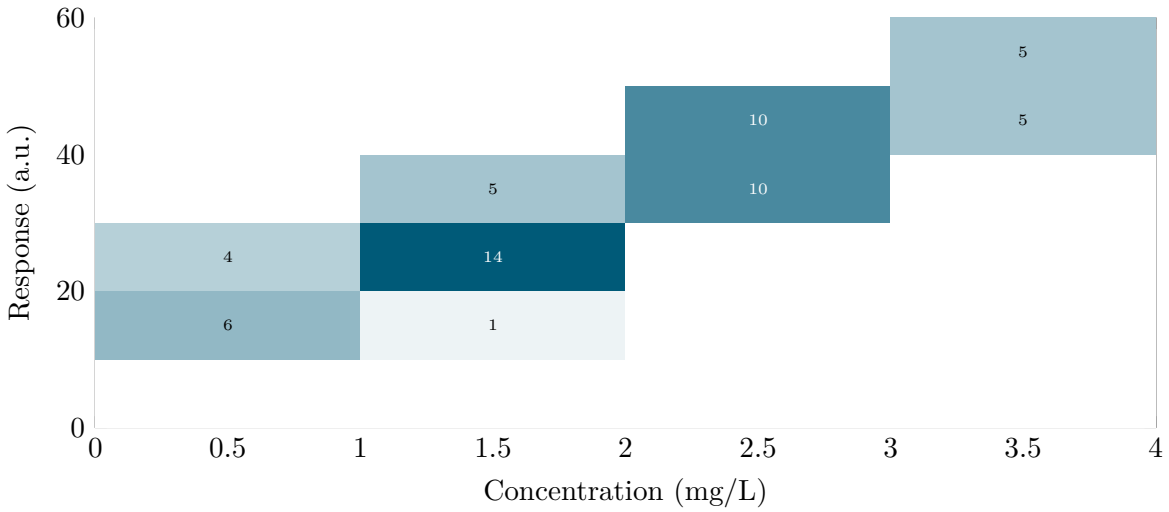


Fig. 20. Adaptive-width histogram at the authored interval settings and density normalization.



The two-dimensional histogram in Fig. 21 extends grouping to pairs of variables and a grid of cells. Its normalization follows the authored count or density convention. The optional R source in Listing 2 reproduces the histogram examples and their settings. Together these views support a consistent question: which proportion of the observed sample occupies a specified region of the measurement space?

Fig. 21. Two-dimensional histogram of the authored paired observations and cell-normalization convention.



Cell labels give observation counts.

Listing 2. Optional R source for the histogram laboratory, including the authored data and bin choices.

```
if (!requireNamespace("ggplot2", quietly = TRUE)) {
  if (requireNamespace("webr", quietly = TRUE)) webr::install("ggplot2")
  else stop("Install ggplot2 first")
}
library(ggplot2)
x <- c(
  36.203861904595, 36.968344965996, 37.373171342811, 37.663249348955, 37.894944427129,
   ↪  38.090935235275, 38.262750520860, 38.417100222768, 38.558256085355, 38.689118543731,
  38.811755599904, 38.927699911254, 39.038123973652, 39.143949102676, 39.245916271163,
   ↪  39.344633882841, 39.440611031230, 39.534281313527, 39.626020324213, 39.716158820762,
  39.804992871389, 39.892791871197, 39.979805043931, 40.066266872452, 40.152401787581,
   ↪  40.238428371541, 40.324563286669, 40.411025115191, 40.498038287925, 40.585837287733,
  40.674671338359, 40.764809834908, 40.856548845594, 40.950219127892, 41.046196276280,
   ↪  41.144913887958, 41.246881056446, 41.352706185469, 41.463130247867, 41.579074559217,
  41.701711615390, 41.832574073766, 41.973729936353, 42.128079638261, 42.299894923847,
   ↪  42.495885731992, 42.727580810167, 43.017658816310, 43.422485193125, 44.186968254526,
  55.813031745474, 56.577514806875, 56.982341183690, 57.272419189833, 57.504114268008,
   ↪  57.700105076153, 57.871920361739, 58.026270063647, 58.167425926234, 58.298288384610,
  58.420925440783, 58.536869752133, 58.647293814531, 58.753118943554, 58.855086112042,
   ↪  58.953803723720, 59.049780872108, 59.143451154406, 59.235190165092, 59.325328661641,
  59.414162712267, 59.501961712075, 59.588974884809, 59.675436713331, 59.761571628459,
   ↪  59.847598212419, 59.933733127548, 60.020194956069, 60.107208128803, 60.195007128611,
  60.283841179238, 60.373979675787, 60.465718686473, 60.559388968770, 60.655366117159,
   ↪  60.754083728837, 60.856050897324, 60.961876026348, 61.072300088746, 61.188244400096,
  61.310881456269, 61.441743914645, 61.582899777232, 61.737249479140, 61.909064764725,
   ↪  62.105055572871, 62.336750651045, 62.626828657189, 63.031655034004, 63.796138095405
)
bin_width <- 4
p <- ggplot(data.frame(x), aes(x)) +
  geom_histogram(breaks = seq(30, 70, by = bin_width), closed = "left") +
  labs(x = "Synthetic detector response (a.u.)", y = "Count") +
  theme_minimal(base_size = 16)
print(p)
b <- ggplot_build(p)$data[[1]]
b[c("xmin", "xmax", "count", "density")]
sum(b$count)
```

geom_histogram(breaks=..., closed="left")

Specify numeric edges and interval closure explicitly, then verify that the bins cover and count every observation.

ggplot_build(p)$data[[1]]

Inspect xmin/xmax, count and density; verify sum(count)=length(x).

ggplot(..., aes(x))

ggplot() creates the plot; **aes(x)** maps the observation column to the horizontal axis. The histogram layer computes counts.

c(...)

Combine the supplied responses or the selected output-column names into a vector.

data.frame(x)

Put the response vector into a named table column for ggplot.

seq(30, 70, by=bin_width)

Generate **explicit bin edges**. Changing bin_width changes these edges, not the observations.

sum(b$count)

Add all bin counts. **Check against 100** so observations do not silently disappear outside the edges.

requireNamespace(..., quietly=TRUE)

Check package availability without attaching it; the logical result selects the setup branch.

webr::install("ggplot2")

Download ggplot2 into browser R if missing. **Network required**, not an upload of learner observations.

stop(...)

Stop with an explanatory setup error. Desktop R needs ggplot2 installed separately.

library(ggplot2)

Attach the installed plotting package so its functions are available by name.

print(p)

Draw the histogram explicitly in the browser R runtime.

labs(x=..., y=...)

Label the measurement axis with its unit and the vertical axis as **Count**.

theme_minimal(base_size=16)

Set the plot theme and font size; the bin counts do not change.

<-, $, [, [[, +, if / else, ! and ::

<- assigns; **$** selects a column; [subsets; [[extracts a list element. + adds plot layers; **if/else** branches; **!** negates; **::** selects a package function.

6 An empirical CDF answers cumulative threshold questions directly

Many analytical questions ask what fraction of observations is at or below a concentration threshold. The empirical cumulative distribution function in Eq. 10 calculates that fraction for every threshold. Each observed value adds its share to the cumulative total, and tied values produce a combined jump. The definition specifies the inclusive threshold convention.

Eq. 10. Empirical CDF with the inclusive threshold, indicator, sample-size and tie conventions.

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{x_i \leq x\}$$

Symbols
$F_n(x)$ and x $F_n(x)$: unitless sample fraction at or below threshold x. The threshold x has the measurement unit.
x_i, n, i and Σ x_i: observed value i; n: sample size; i: index 1 to n; Σ: sum over every observation, including ties.
$\mathbf{1\{x_i \leq x\}}$ Indicator: 1 when $x_i \leq x$, otherwise 0. Equality is included: tied values enter together. Dividing the count by n gives the fraction.

The small worked example in Fig. 22 makes the counting operation visible. The river example in Fig. 23 gives it an analytical interpretation: read horizontally from a concentration to obtain the observed fraction, or start from a cumulative level to locate an empirical threshold under the stated quantile convention. The observations remain the basis for both readings.

Fig. 22. Worked empirical cumulative steps for the authored small sample.

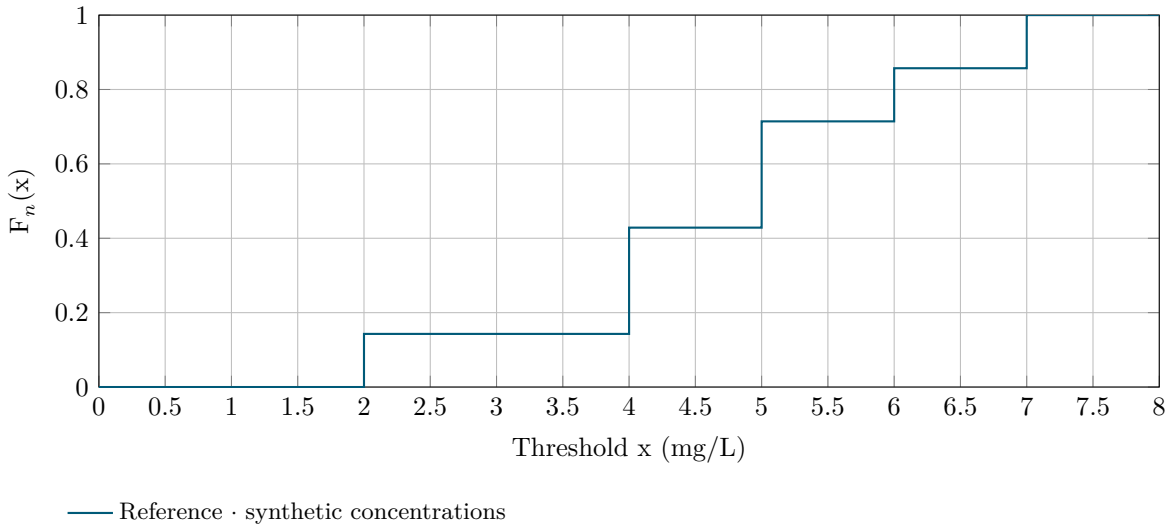
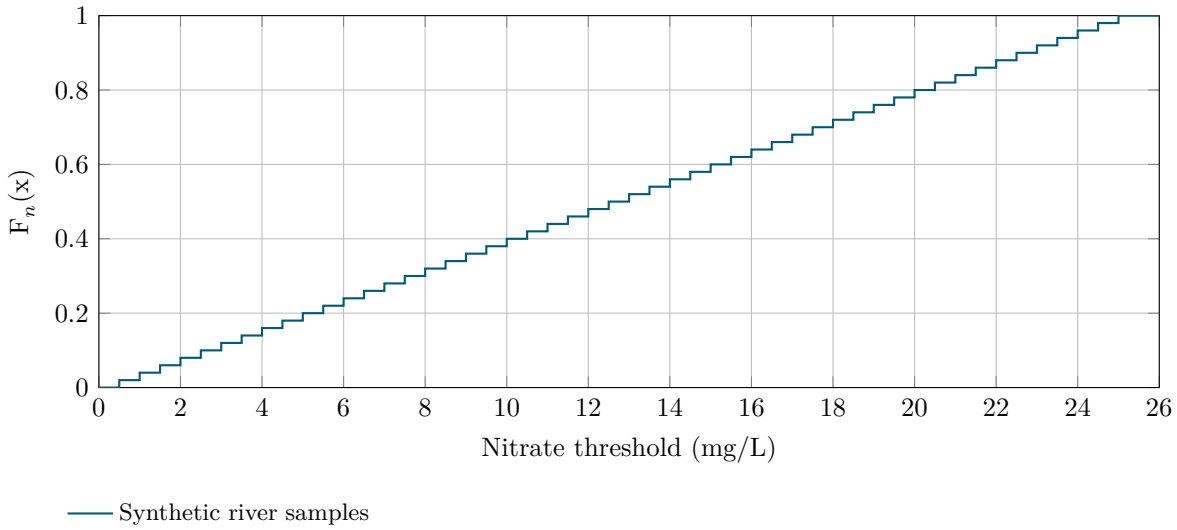
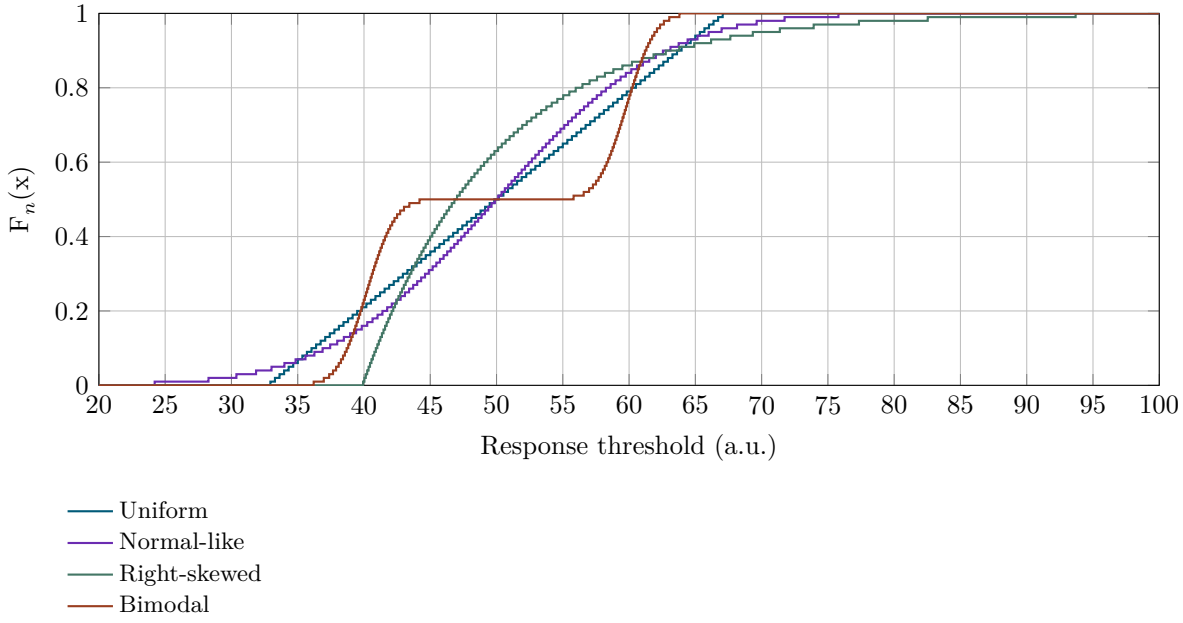


Fig. 23. River-concentration ECDF for cumulative-threshold and empirical-quantile readings.



The shape comparison in Fig. 24 uses cumulative curves to compare complete empirical distributions. A curve lying higher at a fixed concentration represents a larger observed fraction below that concentration. Crossing curves show that the ordering depends on the threshold. Generalizing these empirical comparisons to populations additionally requires an inferential model and a sampling design appropriate to the question.

Fig. 24. Empirical cumulative comparison of the authored distribution shapes, including threshold-dependent curve orderings.



7 Moment summaries describe complementary aspects of shape

Moments summarize a distribution through powers of its values or centred departures. The authored mean calculation in Eq. 11 and comparison matrix in Fig. 25 concern location. The variance calculation and matrix (Eq. 12, Fig. 26) concern squared distance from the centre. Their finite-sample conventions are stated in the attached legends.

Eq. 11. Authored mean calculation and its observation, unit and averaging conventions.

$$\mu = \mathrm{E}[X]$$
$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Symbols
X, $\mathbf{x}_i$, i and n X: random measurement variable; x_i: observed value i; i: index 1 to n; n: sample size.
μ **and** $\bar{x}$ μ: population mean; $\bar{x}$: mean of this sample. Both use the measurement unit; an observed $\bar{x}$ need not equal μ.
E[ ], Σ and 1/n E[ ]: probability-weighted population expectation, requiring a finite first absolute moment. Σ adds observations i = 1 to n; 1/n forms their empirical average.

Fig. 25. Authored distribution comparisons organized by their mean summaries.

Horizontal axes: Response (a.u.).



Eq. 12. Authored variance calculation with its explicit finite-sample denominator and squared units.

$$\sigma^2 = \mathrm{E}[(X-\mu)^2], \quad \mu = \mathrm{E}[X]$$
$$m_2 = \frac{1}{n}\sum_i (x_i - \bar{x})^2 = \frac{n-1}{n} s^2$$

Symbols

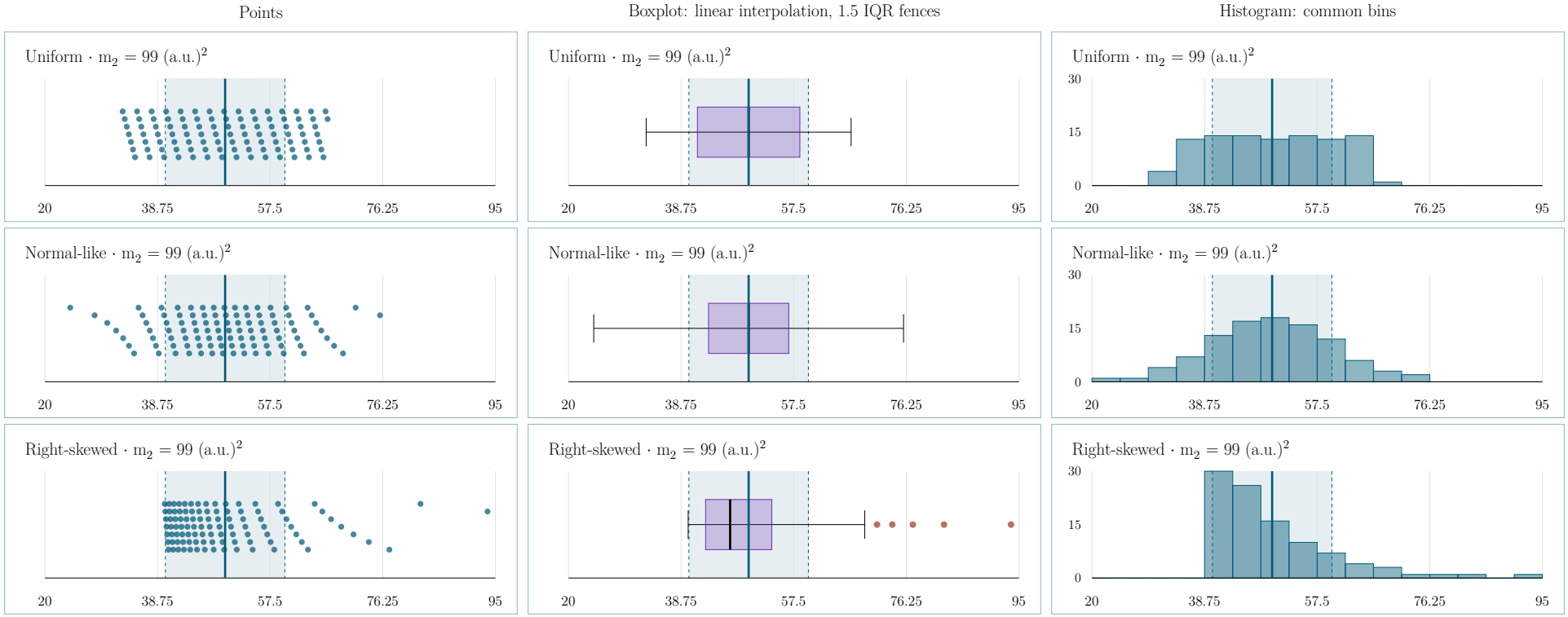
X, μ, E[ ] and σ^2 X: random measurement; μ: its expectation; E[ ]: population expectation; σ^2: population variance. A finite second moment is required.

m_2 and s^2 m_2: empirical variance with divisor n; s^2: sample variance with n−1. Their relation requires $n > 1$. All variances have squared measurement units.

x_i, $\bar{x}$, n, i and Σ x_i: observed value i; $\bar{x}$: sample mean; n: sample size; i: index 1 to n; Σ: sum over those observations. The empirical squared deviations concern this sample, not a known population σ^2.

Fig. 26. Authored distribution comparisons organized by their variance summaries.

Horizontal axes: Response (a.u.).



Standardized third-order departures describe the signed asymmetry represented by the chosen skewness statistic (Eq. 13, Fig. 27). Standardized fourth-order departures emphasize large magnitudes relative to scale (Eq. 14, Fig. 28). The authored convention distinguishes kurtosis from excess kurtosis and specifies the denominator and sample correction used. These conventions matter when comparing software outputs.

Eq. 13. Authored standardized third-moment statistic with its complete centring, scaling and correction conventions.

$$\gamma_1 = \frac{\mathrm{E}[(X-\mu)^3]}{\sigma^3}, \quad \sigma > 0$$
$$g_1 = \frac{m_3}{m_2^{3/2}}, \quad m_2 > 0$$

Symbols

γ_1 **and** $\mathbf{g}_1$ γ_1: population skewness; g_1: descriptive sample counterpart. Both are dimensionless; no small-sample bias adjustment is applied to g_1.

E[ ], X, μ **and** σ E[ ]: population expectation; X: random measurement; μ: population mean; σ: population SD. A finite third absolute moment and $\sigma > 0$ are required.

$\mathbf{m}_3$ **and** $\mathbf{m}_2$ m_3: average cubed deviation from the sample mean; m_2: average squared deviation, each using n. Dividing by m_2^(3/2) removes the cubed unit. g_1 is undefined if $m_2 = 0$.

Fig. 27. Authored shape comparisons accompanying the skewness calculations.

Horizontal axes: Response (a.u.).



Eq. 14. Authored fourth-moment statistic with the kurtosis or excess-kurtosis convention stated explicitly.

$$\kappa = \frac{\mathrm{E}[(X-\mu)^4]}{\sigma^4}, \quad \sigma > 0$$
$$K = \frac{m_4}{m_2^2}, \quad g_2 = K - 3, \quad m_2 > 0$$

Symbols

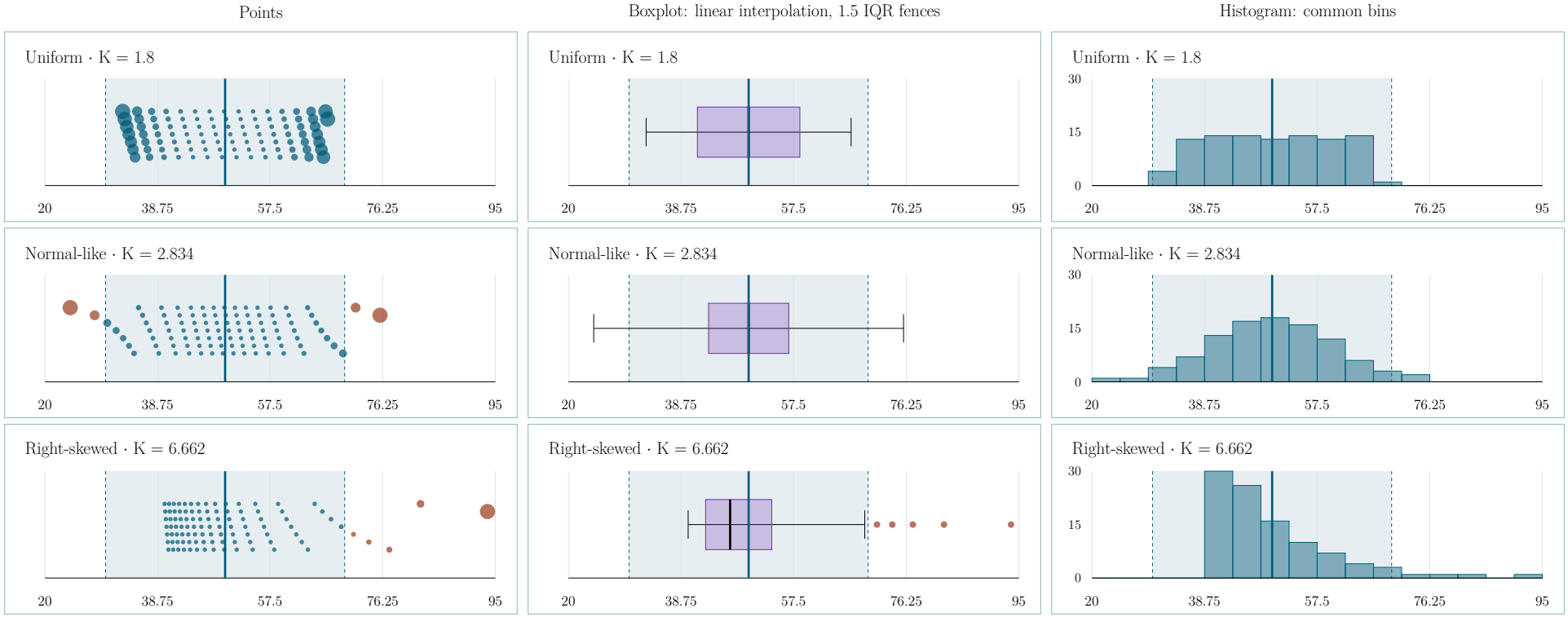
κ, X, μ, σ and E[ ] κ: population kurtosis; X: random measurement; μ: population mean; σ: population SD; E[ ]: expectation. A finite fourth moment and $\sigma > 0$ are required.

K, $\mathbf{m}_4$ and $\mathbf{m}_2$ K: descriptive sample kurtosis. m_4 and m_2 average fourth and squared deviations from the sample mean, using n. K is dimensionless and requires $\mathrm{m}_2 > 0$.

$\mathbf{g}_2$ and 3 g_2: sample excess K−3, without bias correction. Population excess: κ−3; 3 is normal-population kurtosis, not every normal sample's value.

Fig. 28. Authored shape comparisons accompanying the fourth-moment calculations.

Horizontal axes: Response (a.u.).



A moment is one summary of the observations. Interpreting it alongside a dot plot, histogram or ECDF reveals which observations and patterns produce its value. A zero third moment, for example, expresses cancellation of the signed cubic contributions; symmetry is a stronger structural property to examine in the full distribution. The optional source in Listing 3 reproduces the comparisons. These empirical descriptions prepare the transition to probability models, where a distribution's assumptions and parameters make statements about future or repeated observations.

Listing 3. Optional R reproduction of the moment calculations and distribution comparisons.

```
# Deterministic synthetic grids; same construction as the lecture
p <- ((1:100) - 0.5) / 100
standardize <- function(z) 50 + 10 * as.numeric(scale(z))
series <- list(Uniform=standardize(p), Normal=standardize(qnorm(p)), Skewed=standardize(qexp(p)))
offset <- 0
scale_factor <- 1
extreme <- FALSE
metric <- 'variance'  # mean, variance, kurtosis, skewness
stopifnot(metric %in% c('mean','variance','kurtosis','skewness'))
series <- lapply(series, function(x) offset + scale_factor*x)
if (extreme) series$Normal[100] <- series$Normal[100] + 50

describe <- function(x) {
  d <- x-mean(x); m2 <- mean(d^2)
  c(mean=mean(x), m2=m2, sample_variance=var(x),
    kurtosis=if(m2>0) mean(d^4)/m2^2 else NA_real_,
    skewness=if(m2>0) mean(d^3)/m2^1.5 else NA_real_)
}
print(t(sapply(series, describe)))
x <- series$Normal; m2 <- mean((x-mean(x))^2)
cat('n =',length(x),'; sample variance =',var(x),'; m2 =',m2,'\n')

# Shared measurement axes and histogram edges
limits <- range(unlist(series)); if(diff(limits)==0) limits <- limits+c(-1,1)
edges <- seq(limits[1], limits[2], length.out=17)
mark <- function(x) {
  mu <- mean(x); root_m2 <- sqrt(mean((x-mu)^2))
  abline(v=mu,col='deepskyblue3',lwd=2)
  if(metric=='variance') abline(v=mu+c(-1,1)*root_m2,col='deepskyblue3',lty=3)
  if(metric=='kurtosis') abline(v=mu+c(-2,2)*root_m2,col='orange3',lty=3)
  if(metric=='skewness') abline(v=median(x),col='purple3',lty=2,lwd=2)
}
interpolated_box <- function(x) {
  q <- quantile(x,c(.25,.5,.75),type=7)
  width <- q[3]-q[1]
  inside <- x[x>=q[1]-1.5*width & x<=q[3]+1.5*width]
  flagged <- x[x<q[1]-1.5*width | x>q[3]+1.5*width]
  plot(NA,xlim=limits,ylim=c(0,1),yaxt='n',ylab='',
       main='Boxplot: linear interpolation',xlab='Response (a.u.)')
  segments(min(inside),.5,max(inside),.5)
  segments(c(min(inside),max(inside)),.4,c(min(inside),max(inside)),.6)
  rect(q[1],.3,q[3],.7,col='lavender',border='purple3')
  segments(q[2],.3,q[2],.7,lwd=2)
  points(flagged,rep(.5,length(flagged)),pch=16,cex=.5,col='orange3')
  mark(x)
  print(c(Q1=q[1],median=q[2],Q3=q[3],
          lower_whisker=min(inside),upper_whisker=max(inside)))
}
par(mfrow=c(3,3),mar=c(3,3,2.8,0.7),cex=0.7)
for(name in names(series)) {
  x <- series[[name]]; result <- describe(x)
  key <- if(metric=='variance') 'm2' else metric
  label <- paste(name,key,'=',round(result[key],3))
  stripchart(x,method='stack',pch=16,cex=0.45,xlim=limits,
             main=label,xlab='Response (a.u.)'); mark(x)
```

```
  interpolated_box(x)
  hist(x,breaks=edges,right=FALSE,include.lowest=TRUE,
       main='Histogram: common edges',xlab='Response (a.u.)'); mark(x)
}
cat('Variance guide: mean +/- sqrt(m2); kurtosis guide: mean +/- 2*sqrt(m2). Neither is a universal
↪  coverage interval.\n')
a <- rep(c(-3,-1,2),c(5,25,20))
cat('Counterexample: mean =',mean(a),'; m3 =',mean((a-mean(a))^3),'\n')
stopifnot(length(x)==100,isTRUE(all.equal(var(x),mean((x-mean(x))^2)*100/99)))
```

standardize(z)

Local helper: rescale a synthetic grid to **sample mean 50 and sample SD 10**. Not a fitted population model.

scale(z)

By default subtract the mean and divide by **sample SD**. Returns a matrix with scaling attributes.

as.numeric(...)

Convert the scaled matrix to a plain numeric vector, dropping its matrix shape and attributes.

qnorm(p)

Return standard-normal quantiles at p. Here p is a **deterministic midpoint grid**, not random sampling.

qexp(p)

Return exponential quantiles (default rate 1), supplying the right-skewed grid before standardization.

c(...)

Combine values or named results into a vector; does not sort them.

list(...)

Collect the three named series, retaining each as a separate numeric vector.

lapply(series, function(x) ...)

Transform every series by the same offset/scale setting; return a **list** of transformed vectors.

sapply(series, describe)

Run describe() on each series; simplify equally sized result vectors into a matrix.

t(...)

Transpose the summary matrix so each printed **row is one series**.

describe(x)

Local helper: return mean, two variance conventions, skewness and **Pearson kurtosis K, not excess**. Constant data give undefined shape coefficients.

mean(x)

Calculate an arithmetic mean. Averaging powers of centred deviations gives **empirical moments with divisor n**.

var(x)

Calculate **sample variance with divisor n−1**, not descriptive variance with divisor n.

median(x)

Return the middle ordered value, averaging the two middle values for even n. Used for the dashed comparison line.

sqrt(...)

Take the nonnegative square root: convert descriptive variance into a spread guide in the **original unit**.

length(x)

Count vector elements, including ties. Each teaching series should contain 100 observations.

print(...)

Display computed result vectors or matrices explicitly in the text output.

cat(...)

Write labelled values and line breaks to the text output; does not return a table.

unlist(series)

Flatten the list **only to find shared axis limits**; series stay separate for analysis.

range(...)

Return the smallest and largest values across the series, defining shared measurement limits.

diff(limits)

Subtract consecutive values. For two limits this gives the axis width; zero triggers safe plotting expansion.

seq(..., length.out=17)

Generate 17 shared edges spanning the limits, giving **16 histogram bins** in every row.

quantile(x, ..., type=7)

Compute quartiles by **linear interpolation**, not the alternative hinges of some boxplot implementations. The numeric argument selects this calculation in R.

min(inside), max(inside)

min() selects the smallest and **max()** the largest observation inside the fences. Those observations end the whiskers.

rep(...)

Repeat point heights or values. A vector of counts sets each value's multiplicity in the asymmetric counterexample.

par(mfrow=c(3,3), ...)

mfrow fills nine panels row by row; **mar** sets margins; **cex** sets text/symbol scaling.

names(series)

Return the series names for iteration and labels.

paste(...)

Combine series name, descriptor name and value into a plot title.

round(..., 3)

Round the **displayed label** to three decimals; calculations retain unrounded values.

stripchart(..., method="stack")

Draw individual observations with stacking. Point height is **not time or a second measurement**.

plot(NA, xlim=..., ylim=...)

Create an empty frame for the custom boxplot, with explicit shared limits and axis labels.

segments(...)

Draw line segments for whiskers, end caps and the median.

rect(...)

Draw the box between the explicitly calculated first and third quartiles.

points(...)

Add orange points for observations outside the fences; they remain in the sample.

abline(v=..., col=..., lty=..., lwd=...)

Add vertical reference lines: **v** position, **col** colour, **lty** dash style, **lwd** width. No probability is estimated.

hist(..., breaks=edges, right=FALSE, include.lowest=TRUE)

Use shared explicit bins. **right=FALSE** makes them left-closed; **include.lowest=TRUE** includes the terminal endpoint for this closure.

mark(x)

Local helper: draw the mean and metric-specific guides. **metric** selects the guide, not a new dataset or confidence interval.

interpolated_box(x)

Local helper: draw linearly interpolated quartiles, observed whiskers and flagged points; print their numerical boundaries.

stopifnot(...)

Stop if a check fails. The metric check limits allowed settings; the final check inspects the **last loop series**, not all three.

isTRUE(all.equal(...))

all.equal() compares with numerical tolerance, returning TRUE or a diagnostic; **isTRUE()** produces one logical pass/fail value.

function, for, if / else, <-, $, [, [[, %in% and arithmetic

function defines helpers; **for** repeats; **if/else** branches. **<-** assigns; **$**, [, [[select data; **%in%** tests membership. **:** makes integer sequences; $+ - * /$ calculate; $\hat{}$ raises powers. Comparisons return logical values; **& / |** combine them. **NA_real_** marks undefined numeric coefficients.